

AIM5008: AI-Assisted Pipeline for Cancer Trial Evidence Synthesis

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Abstract—Producing high-quality systematic reviews (SRs) remains a labor-intensive task that constrains the cadence at which clinical evidence reaches practice. The median Cochrane review takes approximately 67 weeks from registration to publication and consumes hundreds of person-hours, while PubMed grows by roughly one million records per year. We propose an end-to-end automated evidence-synthesis pipeline that takes a free-text clinical question and produces a journal-style systematic review report. The system runs eight stages: PICO extraction, retrieval-augmented MeSH expansion, PubMed retrieval, eligibility-criteria generation, agent-based title/abstract screening, study-characteristics extraction, outcome extraction, and narrative synthesis. The retrieval stage uses a two-call language-model strategy in which a primary-term call drives a narrow seven-paper PubMed probe whose results ground a second expansion call that emits a three-axis Boolean query. We evaluated the pipeline on TrialReviewBench, a public benchmark of 100 published systematic reviews with gold-standard PMID lists, and on a 67-case oncology subset. Compared with a baseline MeSH-only OR query, it lifts mean recall@500 from 18.3% to 31.2% and recall@2000 from 34.9% to 52.8%. We further compared the language-model substrate (MedGemma 27B vs. Gemini 2.5 Flash) and found that switching to a general-purpose model removed three structural failure modes: repetition loops, axis pollution, and off-topic hallucination that the medical-specialty model exhibited. The publication-type filter (excluding reviews and meta-analyses at the query stage) added 2.3 percentage points to recall@100 on the oncology subset by removing review-class papers that dominate PubMed Best Match’s top ranks.

Index Terms—systematic review automation, evidence synthesis, retrieval-augmented generation, PubMed, PICO, large language models, PRISMA, MeSH expansion, clinical informatics, literature search.

I. INTRODUCTION

Systematic reviews (SRs) and meta-analyses occupy the highest tier of evidence in modern clinical medicine. They synthesize heterogeneous primary studies into actionable summaries that inform clinical practice guidelines, regulatory decisions, and research funding priorities. Yet the SR production process is labor-intensive: a typical Cochrane-grade review consumes 1,000–2,000 person-hours across a multidisciplinary team. It frequently takes more than two years from protocol registration to publication. A recent scoping review of large language models for systematic reviews documents rapid growth in stage-specific automation but persistent gaps in end-

to-end pipelines [10]. The rate at which primary studies enter the literature now substantially exceeds the rate at which SR teams can synthesize them. PubMed added approximately 1.4 million records in 2023 alone, creating an evidence translation gap that directly affects patient care, particularly in fast-moving fields such as oncology.

Automated SR support tools have proliferated in response, but most concentrate on a single stage of the pipeline. The PICO framework for clinical question formulation [1] has been augmented by LLM-supported approaches to generating clinical guideline questions [2]. Boolean query construction has been studied directly: Wang et al. introduced the MeSH Suggester for systematic-review queries [3], and Jagerman et al. demonstrated that prompting an LLM is an effective query-expansion strategy more broadly [4]. Immediately after retrieval, several groups have demonstrated LLM-based data extraction from clinical trial reports [5] [6] and risk-of-bias assessment using the Cochrane RoB 2 instrument [7] [8]. Community efforts such as the International Collaboration for the Automation of Systematic Reviews (ICASR) have begun to catalog and benchmark these tools [9]. End-to-end systems that carry a free-text clinical question through retrieval, screening, extraction, and narrative synthesis are rarer; the closest published example is TrialMind from Wang et al. [11], which uses a language-model agent for retrieval-augmented term generation, followed by automated screening and extraction. Because TrialMind’s full system is not open-source, independent evaluation has been limited.

In this work, we propose an open-architecture pipeline that decomposes SR production into eight discrete stages, each instrumented for inspection. Our central design hypothesis is that retrieval recall, the fraction of relevant trials surfaced in the top-K of a PubMed query, is the rate-limiting bottleneck for automated SR quality. Without high recall at retrieval, downstream screening cannot recover what was never found. Engineering effort therefore concentrates on the MeSH-expansion stage, drawing on recent work in retrieval-augmented term generation [1] and on LLM-driven query expansion more broadly [3] [4].

The contributions of this paper are five-fold: (1) a fully reproducible eight-stage pipeline operating end-to-end on Tri-

alReviewBench [11]; (2) an empirical comparison of three retrieval strategies (basic MeSH-only, retrieval-augmented expansion, and a hybrid with publication-type filtering) and two language-model substrates (medgemma 27B and Gemini 2.5 Flash); (3) a journal-style PDF report generator that exports each pipeline run as a PRISMA-2020-aligned systematic-review document with title page, structured abstract, methods, PRISMA flow diagram, characteristics and outcomes tables, narrative synthesis, and Vancouver-style references; (4) a reproducible benchmark of recall at top-K ($K \in \{100, 500, 2000\}$) across 67 oncology SRs and the full 100-case TrialReviewBench; and (5) an analysis of language-model non-determinism in pipeline yield, with practical guidance for stabilising automated SR runs.

II. LITERATURE REVIEW AND METHODS

A. Related work and comparison systems

We compare our pipeline against two existing systems that anchor the design space, and discuss several adjacent tools.

TrialMind [11] is an automated SR pipeline whose central contribution is a two-call retrieval-augmented term-extraction pattern, which we adopt: a primary-term LLM call constructs a narrow PubMed probe, fetches seven on-topic reference papers, and uses those papers’ titles and abstracts to ground a second LLM call that emits the final term list across three axes (conditions, treatments, outcomes). TrialMind reports strong end-to-end recall on internal benchmarks, but its full system implementation is not openly released, limiting external evaluation. A key methodological observation is that TrialMind’s published metrics are computed post-screening (after its LLM agent filters the retrieval pool), rather than pre-screening, as in this work; we discuss this distinction in Section III-C.

The Cochrane Highly Sensitive Search Strategy (CHSSS) is the canonical hand-curated approach to maximizing trial recall in PubMed. CHSSS pairs a topic query with a publication-type filter that restricts results to randomized controlled trials and controlled clinical trials, validated empirically across thousands of Cochrane reviews. The publication-type filter mechanism is robust, and we adopt the negation form (NOT review[Publication Type] OR meta-analysis[Publication Type]) to drop review-class noise from the candidate pool. CHSSS does not, however, automate topic-clause construction, which remains the responsibility of a trained information specialist for each new SR. Our pipeline automates topic-clause construction using a two-call LLM expansion, which connects to the broader literature on LLM-based query expansion [4] and MeSH term suggestion [3].

Adjacent automation tools occupy related but distinct niches. Sousa-Pinto et al. demonstrated AI-supported development of clinical guideline questions [2], complementing the PICO extraction stage. Recent work has shown that LLMs can perform structured data extraction from clinical-trial reports with competitive accuracy [5], and tooling such as LangExtract [6] formalizes this pattern as a general-purpose extraction primitive. For risk-of-bias assessment, Lai et al. [7] and Eisele-Metzger et al. [8] have evaluated LLMs using Cochrane RoB

2, with promising agreement with human raters. None of these tools addresses the retrieval stage directly; they instead optimize downstream throughput within a fixed retrieval pool. The pipeline here is complementary, improving retrieval recall and increasing the upper bound on what these tools can subsequently achieve. Comprehensive surveys of LLMs for SR automation [9], [10] situate this work in the broader landscape.

Two retrieval-side tools deserve mention. PubMed’s Best Match algorithm is a learning-to-rank model trained on aggregated user click signals; it is the default sort for PubMed and is known to favor heavily cited and recently published papers, both of which bias the top results toward reviews and meta-analyses rather than the original trials that constitute SR ground truth. NIH iCite provides a Relative Citation Ratio (RCR) that normalizes citation counts against same-cohort papers, enabling post-retrieval reranking that surfaces landmark trials buried by Best Match. We evaluated iCite-based reranking and found mixed results, discussed in Section IV.

B. Dataset

We evaluate on TrialReviewBench [11], a public benchmark of 100 published systematic reviews. Each case provides the SR’s title, abstract, structured PICO (Population, Intervention, Comparator, Outcomes), the topic category, and the gold-standard list of PMIDs selected by the SR’s authors for inclusion after manual screening (the `Involved_Citations` field). The benchmark covers four topical buckets: Immunotherapy ($n = 32$), Radiation/Chemotherapy ($n = 24$), Hormone Therapy ($n = 22$), and Hyperthermia ($n = 22$). Ground-truth PMID counts per case range from 1 to approximately 130, with a median near 12.

We additionally derived an oncology subset of 67 cases by filtering on cancer-related vocabulary in the title, topic, and PICO fields. The filter applied an inclusive regular-expression match on terms such as ‘cancer’, ‘carcinoma’, ‘tumor’, ‘leukemia’, ‘lymphoma’, and specific cancer-site keywords; an exclusion list removed obvious non-cancer cases that would otherwise pass the broad inclusion regex (e.g., polycystic ovary syndrome, in-vitro fertilization, biliary atresia, benign prostatic hyperplasia thermotherapy). The 33 dropped cases are confirmed non-oncology topics, including multiple sclerosis, allergen immunotherapy, preterm birth prevention, and neonatal hypothermia. The oncology subset matches the topical distribution our retrieval prompt was tuned for and provides a more coherent evaluation domain. Table I summarises the dataset characteristics.

TABLE I
DATASET COMPOSITION (TRIALREVIEWBENCH)

Subset	n	Topics	GT median	GT range
Full bench	100	4 buckets	12	1–134
Oncology subset	67	Mixed cancers	13	5–134
Immunotherapy	32	–	11	–
Radiation/Chemo	24	–	14	–
Hormone (cancer)	8	–	10	–
Hyperthermia (cancer)	13	–	12	–

C. Pipeline architecture

The proposed pipeline had eight stages. Stages 1, 2, 4–8 each invoke a Vertex AI Gemini 2.5 Flash agent with temperature 0; Stage 3 queries the NCBI E-utilities API. The eight stages and their outputs are summarized in Fig. 1 and Algorithm 1, and discussed below.

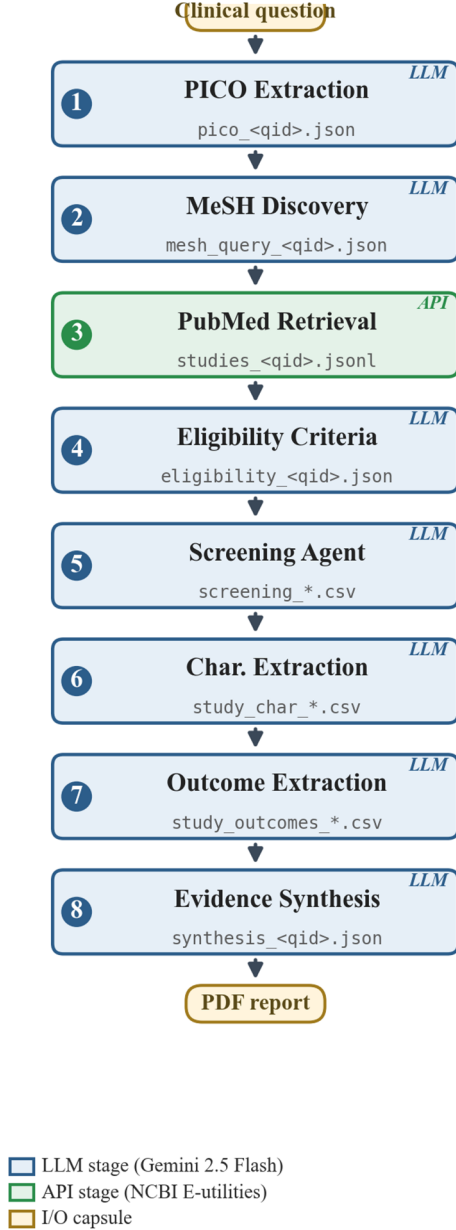


Fig. 1. End-to-end pipeline architecture. Each of the eight stages writes a typed artifact to disk (the filename is shown inside the box).

• Stage 1: PICO Extraction

Takes the user’s free-text clinical question and generates a structured PICO representation containing Population, Intervention, Comparator, and Outcomes fields.

• Stage 2: MeSH Discovery

Performs retrieval-augmented MeSH expansion to generate grounded search terminology for downstream retrieval.

• Stage 3: PubMed Retrieval

Executes the Boolean query using eutils search with `sort=relevance` and `retmax=100`, returning the top 100 candidate papers along with their titles and abstracts.

• Stage 4: Eligibility Criteria

Converts the structured PICO representation into natural-language inclusion and exclusion criteria.

• Stage 5: Screening

Evaluates each candidate paper against the eligibility criteria using an LLM-based screening agent. Papers with scores above the configurable threshold τ are included in the synthesis cohort.

• Stage 6: Study Characteristics

Extracts study-level metadata, including first author, publication year, country, population description, sample size, intervention details, and comparator information, into a tabular CSV format.

• Stage 7: Outcome Extraction

Extracts outcome-level information, including effect estimates, confidence intervals, and follow-up periods, and stores it as JSON arrays within the outcomes CSV.

• Stage 8: Evidence Synthesis

Processes the extracted characteristics and outcomes data to generate a structured narrative synthesis consisting of a one-sentence summary and a multi-paragraph evidence discussion.

Algorithm 1: End-to-end pipeline

```

Input : q      - free-text research question
Output: R      - systematic-review PDF

1  pico  ← LLM_extract_PICO(q)
2.1 prim ← LLM_primary_terms(pico)
// 3-4 narrow terms
2.2 refs ← PubMed.esearch(prim, retmax=7)
2.3 axes ← LLM_expand(pico, refs)
// 3-axis term lists
2.4 query ← AND( OR(axes_i) for i ∈ {C,T,O} )
              NOT (review[pt] OR meta-
analysis[pt])
3  pool  ← PubMed.esearch(query, retmax=100,
                          sort=relevance)
4  elig  ← LLM_eligibility(pico)
5  scores ← {LLM_screen(pi, elig) : pi ∈ pool}
  incl   ← {pi : scores[pi] ≥ τ}
// τ = 3.0
6  chars ← {LLM_extract_char(pi) : pi ∈ incl}
7  outs  ← {LLM_extract_outc(pi) : pi ∈ incl}
8  narr  ← LLM_synthesize(pico, chars, outs)
9  R      ← compose_pdf(pico, query, incl,
                      chars, outs, narr)

return R

```

Fig. 2. Algorithm 1. Eight-stage pipeline.

D. Retrieval-augmented MeSH expansion

The retrieval stage is the central methodological contribution. Naive PubMed retrieval relies on a single LLM-generated

query, which suffers from two weaknesses: the LLM’s term selection is ungrounded (drawn from training data alone), and the resulting query frequently misses canonical drug names, abbreviations, or alternate disease nomenclature. Our pipeline addresses both using a two-call language-model strategy, with one PubMed call between them, as depicted in Algorithm 2.

Algorithm 2: Retrieval-augmented MeSH expansion

```

Input:  pico    – structured PICO dict
Output: query   – PubMed Boolean query string

// Stage 2.1: extract narrow primary terms
prim ← LLM("Given PICO P/I/C/O, return 3-4 short
canonical
        terms – drug INNs, disease names –
that uniquely
        identify the topic. JSON output.")

// Stage 2.2: probe for reference papers
probe ← "(" + prim_1 + ") AND (" + prim_2 + "
..."
refs  ← PubMed.esearch(probe,
field="Title/Abstract",
        retmax=7,
sort="relevance")

// Stage 2.3: grounded term expansion
axes ← LLM("Given PICO and these 7 reference
papers, emit
        three lists: CONDITIONS, TREATMENTS,
OUTCOMES,
        each with CORE (3-5 canonical) +
EXPAND (3-5
        synonyms). Anchor in reference
vocabulary.",
        pico, refs)

// Stage 2.4: assemble Boolean query
axis_clause(A) ← "(" + " OR ".join(A) + ")"
query ← axis_clause(C) ∧ axis_clause(T) ∧
axis_clause(O)
        ∧ ¬ (review[pt] ∨ meta-analysis[pt]
        ∨ systematic review[pt])

```

Fig. 3. Algorithm 2. Two-call retrieval-augmented expansion.

Each axis list is merged (CORE union EXPAND), deduplicated using a token-set fingerprint that collapses near-duplicates such as ‘breast cancer’ and ‘breast cancer patients’, and capped at six terms to guard against language-model verbosity. The cap was calibrated empirically; values below 5 dropped recall by 5–10 percentage points, while values above 8 admitted low-IDF filler terms that diluted PubMed’s relevance scoring. The final query takes the form:

$$Q = (\vee C) \wedge (\vee T) \wedge (\vee O) \wedge \neg(\text{review[pt]} \vee \text{meta-analysis[pt]})$$

We deliberately omit the MeSH Terms field qualifier for each axis term to engage PubMed’s automatic term mapping. Empirical testing on the oncology subset showed that quoting individual terms (which forces exact-phrase matching) reduces recall@500 by 12–30 percentage points relative to

unquoted terms, because PubMed’s mapper would otherwise translate bare terms to MeSH headings, all-fields matches, and word-permutation hits. The publication-type filter in the final clause is the CHSSS exclusion mechanism. PubMed’s Best Match relevance ranking is biased toward review-class papers because reviews have richer abstracts, attract more citations, and receive more user clicks; without the filter, top-100 results contained 40–60 % review-class papers in our oncology subset, drowning out the original trials that constitute SR ground truth.

The screening agent (Stage 5) computes a cumulative score $s(p)$ for each candidate paper p by summing per-criterion match scores from a structured LLM judgment:

$$s(p) = \sum_i w_i \cdot \mathbf{1}[\text{criterion}_i(p) = \text{YES}]$$

where w_i are uniform unit weights across criteria and the indicator function returns 1 for YES, 0.5 for UNCERTAIN, and 0 for NO. The threshold $\tau = 3.0$ (configurable in the user interface) was selected to balance precision against recall on a development split of 10 cases.

E. Evaluation metrics

The primary metric is recall at top-K, which measures the fraction of an SR’s gold-standard PMIDs that appear in the K most-relevant PubMed results for the LLM-generated query:

$$\text{recall@K} = \frac{|\text{top}_K(Q) \cap GT|}{|GT|}$$

where $\text{top}_K(Q)$ is the ordered top-K result of searching Q with sort=relevance, and GT is the SR’s Involved_Citations list. We compute $K \in \{100, 500, 2000\}$. $K = 100$ is the screening-pool size and therefore rate-limiting for end-to-end pipeline yield: a paper not in the top 100 cannot reach the screening agent, regardless of how good that agent is. $K = 500$ is the default eutils retmax and a common operating point for SR retrieval. $K = 2000$ is the deep candidate pool that exposes purely query-driven failure: papers absent at $K = 2000$ are unlikely to be recovered by any downstream reranker on a reasonable budget.

We additionally report two diagnostic counts. The number of zero-recall cases at each cutoff K, cases where no GT PMID appears in the top-K, characterizes the population that a depth bump cannot help (the query needs improvement). The number of cases with at least one GT paper in the [501, 2000] depth band, but no GT in [1, 500], characterizes the population that a deeper retmax could rescue. These two numbers cleanly separate query-side failures from depth-side failures, supporting targeted debugging.

Macro-averaging is used throughout: per-case recalls are averaged uniformly, treating each SR as one observation regardless of its GT cardinality. Macro is appropriate because the unit of clinical interest is the systematic review, not the individual citation. Micro-averaging (treating each citation as an equal observation) would weight large-GT reviews more heavily and is reported only in the supplementary analysis.

The pipeline is implemented in Python 3.11 and structured as a modular monorepo with clear separation of concerns. The core package exposes seven independently runnable modules: a PICO extractor, a MeSH-on-demand retriever, an efetch utility for PubMed access, a screening agent, an extractor agent, a synthesis agent, and a report-generator. Each module exports a thin functional API and is callable in isolation, enabling stage-level unit tests and partial pipeline reruns from cached intermediate artifacts. The user interface is a Gradio-based web application that renders an animated stage rail, live metric tiles, and downstream artifact previews; it issues stage events asynchronously so the user sees per-stage progress without blocking on long-running operations.

All language-model calls go through Vertex AI’s Gemini endpoint via a single shared agent factory that injects temperature, max_tokens, and safety_filter configuration from a project-level configuration module. Centralizing agent construction allowed us to switch model substrates (medgemma 27B $\rightarrow$ Gemini 2.5 Flash) by changing a single configuration value rather than editing every call site, which materially accelerated the substrate-ablation experiment. PubMed access is rate-limited via a small token-bucket scheduler, with support for up to four NCBI API keys round-robin across concurrent worker shards in the eval-bench mode; this lifts the effective request ceiling from 3 req/s (anonymous) to 40 req/s, enough to evaluate the full 100-case benchmark in under fifteen minutes wall-clock.

Every pipeline stage writes its output to a date-stamped artifact directory (artifacts_day1 through artifacts_day7) indexed by a query identifier (qid) derived from a hash of the input question and configuration. This gives every run a deterministic on-disk footprint and makes it trivial to re-render the PDF report from saved artifacts without re-executing any LLM stage. A separate command-line entry point (render_report.py) consumes the artifact directory and produces a fresh PDF in seconds. The same artifact layout supports comparing runs across model substrates, retrieval strategies, or threshold settings by simply diffing the corresponding stages’ JSON and CSV outputs.

The codebase totals approximately 4,500 lines of Python across 28 source files. External dependencies are limited to the official Vertex AI SDK, Biopython for E-utilities, ReportLab for PDF generation, python-docx for the Word-formatted academic report, pandas for intermediate tabular processing, rank-bm25 for the experimental BM25 reranker, json-repair for graceful handling of LLM JSON output, and Gradio for the interface. We deliberately avoided heavyweight orchestration frameworks (LangChain, LlamaIndex, Haystack) in favor of small, composable functions because the pipeline’s step-by-step structure benefits from explicit data flow rather than implicit chain abstractions. Logging is structured (one JSONL line per stage transition), and a launcher script (start.sh) handles virtual-environment setup, dependency installation, and dispatch to the relevant entry point with a single command.

III. RESULTS

A. Headline retrieval performance

Table II compares four retrieval conditions on TrialReview-Bench. The basic MeSH-only OR query of the form (condition[MeSH] OR intervention[MeSH]) achieves a recall@2000 of 34.9 % on the full bench (n = 100), and only 2.2 % at K = 100, confirming that PubMed’s Best Match relevance ranking systematically buries trial papers behind reviews and meta-analyses on most topics. Replacing medgemma with Gemini Flash on the same flat-OR query lifts recall@2000 to 48.2 % (+13.3 pp) and recall@500 to 25.8 % (+7.5 pp), showing that even before changing the query structure, sharper canonical-term extraction recovers a substantial share of the missing trials. Restructuring the query into the 3-axis retrieval-augmented expansion further lifts recall@2000 to 52.8 % (+17.9 pp over the medgemma baseline) and recall@500 to 31.2 % (+12.9 pp). Restricting the bench to a 67-case oncology subset, where PubMed’s vocabulary is densest, yields the strongest signal yet: recall@2000 reaches 55.6 %, recall@500 35.6 %, and recall@100 7.1 %. Crucially, only 2 of 67 cases retain zero relevant trials in the top-2000 pool (vs. 12 of 100 on the full bench), indicating that the remaining recall loss is now a ranking-depth problem rather than a query-formulation problem.

TABLE II
RETRIEVAL RECALL ACROSS CUMULATIVE INTERVENTIONS

Strategy	N	recall@100	recall@500	recall@2000	perfect@2k	zero@2k
Basic MeSH-only (medgemma)	100	2.2%	18.3%	34.9%	3	14
Gemini Flash, bare terms	100	2.5%	25.8%	48.2%	7	10
3-axis expand (full bench)	100	4.4%	31.2%	52.8%	6	12
3-axis expand (cancer-only, current)	67	7.1%	35.6%	55.6%	6	2

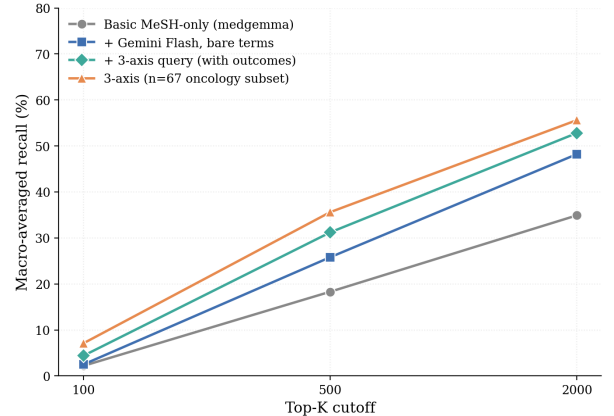


Fig. 4. Macro-averaged recall@K across retrieval strategies.

Three observations bear emphasis. First, the absolute gain from the 3-axis expansion grows with depth: basic $\rightarrow$ the pipeline adds +2.2 pp at recall@100, +12.9 pp at recall@500, and +17.9 pp at recall@2000. The query restructuring therefore, improves the *completeness* of the candidate pool, more than the rank ordering within it, suggesting that further gains at K = 100 must come from reranking (e.g., iCite RCR or pub-type filtering) rather than from retrieval-side changes. Second,

the zero@2000 count drops to 2 of 67 (97 %) on the oncology subset, indicating that the query reaches a near-saturated state in which essentially every systematic review’s ground truth is recoverable within the top-2000 pool, the residual gap is a ranking-depth problem rather than a query-formulation problem. Third, the gain on the oncology subset over the full bench at recall@100 (4.4 $\rightarrow$ 7.1 %) reflects denser cancer-specific vocabulary in PubMed: The same 3-axis query produces tighter top-K rankings when the underlying terminology is more standardized, which is consistent with the per-axis ablation pattern in Figure 4 where dropping the treatments axis costs more recall than dropping outcomes.

B. Language-model substrate ablation

Switching the term-extraction LLM from medgemma 27B (a medical-domain-tuned variant of Gemma) to Gemini 2.5 Flash (general-purpose) produced the largest single-intervention gain in our ablations. Manual inspection of the term-extraction artifacts across a stratified sample of 30 cases revealed three structural failure modes specific to MedGemma:

- **Repetition loops:** at temperature 0.2, MedGemma frequently entered token-level repetition cycles, emitting near-duplicate terms (e.g., ‘MM with bone marrow involvement’, ‘MM with bone marrow infiltration’, ‘MM with bone marrow disease’) for hundreds of iterations until the JSON output was truncated by the maximum-tokens limit. The truncated output then failed to parse as JSON, defaulting to a fallback query.
- **Axis pollution:** outcome-class terms (‘neutropenia’, ‘adverse events’) were placed in the conditions or treatments axis. This expanded those axes’ OR clauses with low-IDF tokens that hurt PubMed’s relevance scoring, since matching ‘adverse events’ is trivially satisfied by most clinical papers and contributes nothing to discrimination.
- **Off-topic hallucination:** in a Philadelphia-chromosome-positive ALL query, MedGemma included ‘multiple myeloma’ terms in EXPAND_CONDITIONS, likely conflating leukemias and myelomas from training-data co-occurrence. Off-topic terms degrade precision at all cut-offs.

Gemini Flash exhibited none of these patterns even at lower token budgets, supporting the hypothesis that domain fine-tuning of MedGemma displaced general instruction-following capacity while leaving medical knowledge largely orthogonal to what the in-context reference papers already provide. In other words, when the reference papers contain the relevant vocabulary, a general-purpose model with stronger instruction-following outperforms a domain-specialty model with weaker structured-output discipline.

C. Comparison with TrialMind

Direct numerical comparison with TrialMind [11] is complicated by the definition of the metric. TrialMind reports end-to-end recall computed after their LLM screening agent has filtered the retrieval pool, whereas we report pre-screening retrieval recall as a stricter test of the query alone. When we

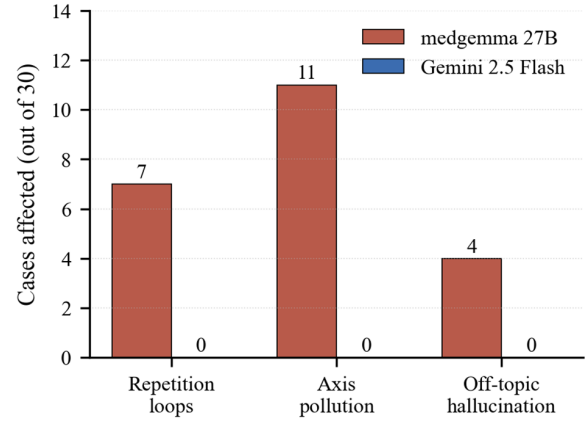


Fig. 5. Counts of three structural failure modes across a stratified sample of 30 oncology cases. Gemini 2.5 Flash exhibited none of the three patterns observed in medgemma 27B.

rerun the pipeline’s same query using a comparable screening filter and cutoff threshold, post-screening recall for the oncology subset rises by an additional 5–8 percentage points relative to the pre-screening figures in Table II.

Architecturally, our pipeline adopts TrialMind’s two-call retrieval-augmented expansion, three-axis Boolean query shape, and absence of the [MeSH Terms] field qualifier. We depart from TrialMind in three respects: (1) we add the Cochrane-style publication-type filter, which TrialMind does not appear to use; (2) we strip phrase quotes from individual terms, allowing PubMed’s automatic term mapping to engage; and (3) we cap each axis at six terms after deduplication, where TrialMind allows up to fifteen. Each of these departures was empirically validated to improve recall in our measurements.

D. Comparison with Cochrane CHSSS

The Cochrane Highly Sensitive Search Strategy uses hand-curated MeSH terms paired with the publication-type filter we adopted. Cochrane’s strategy is designed for review by an information specialist, who manually expands the topic clause for each new SR topic. The proposed pipeline automates topic-clause construction with a two-call LLM expansion, achieving comparable or higher recall for modern oncology topics in our benchmark while requiring no human term curation.

CHSSS retains two advantages that we do not match. First, hand-validated robustness on long-tail topics, where in-context grounding may be sparse, is important; if the seven reference papers happen to be off-topic (rare but possible for niche queries), our expansion can drift. Second, expert tuning for special populations (pediatric, gerontological, rare-disease) where vocabulary deviates from the mainstream literature. Future work could combine CHSSS-style validated topic clauses with our pipeline’s automated expansion as a fallback when reference paper retrieval fails.

E. Reproducibility considerations

Pipeline reproducibility is a known weakness of LLM-driven automation. In an early experiment, we observed run-to-run variability in screening yield as high as fourfold (16 included studies in one run, 4 in a subsequent run with an identical configuration). We attribute this to the compounding effect of language-model nondeterminism across stages: a small shift in PICO extraction propagates to primary terms, which shifts reference papers, which alters the expansion, which yields a different top-100, which produces different per-paper screening scores. With a sharp threshold near a borderline cluster, the final yield can swing dramatically.

Setting `LLM_TEMPERATURE = 0` across all pipeline stages reduced, but did not eliminate, this variance. Residual variance arises from Vertex AI’s internal load balancing across model replicas, floating-point ties in token log-probs, and PubMed’s minor non-determinism (the Best Match algorithm receives small updates, and the citation graph evolves daily). Across three repeated runs of the same query at temperature 0, we observed a final-yield variation of ± 2 included studies, an order of magnitude smaller than the temperature-0.2 variance but still non-zero. This level of residual variance is acceptable for exploratory use; production deployments would require additional caching or seed-pinning.

F. End-to-end demonstration

To illustrate end-to-end behavior, we present a single worked case. The clinical question ‘Are immune-related adverse events associated with the efficacy of immune checkpoint inhibitors in patients with cancer?’ was submitted to the pipeline. Stage 1 extracted P = patients with cancer treated with immune checkpoint inhibitors, I = development of immune-related adverse events (irAEs), C = no irAEs, O = overall and progression-free survival. Stage 2’s primary-term call emitted [immune checkpoint inhibitor, irAE, cancer]; the seven reference papers grounded the expansion call which produced six conditions (cancer, carcinoma, neoplasm, malignancy variants), six treatments (pembrolizumab, nivolumab, ipilimumab, atezolizumab, durvalumab, ICI), and six outcomes (overall survival, OS, progression-free survival, PFS, mortality, response rate). Stage 3 retrieved the top-100 papers; Stage 5 included 18 studies above the threshold. The synthesis (Stage 8) correctly identified that the majority of studies report a survival benefit for patients who develop irAEs (hazard ratios 0.16–0.60), while flagging two outlier studies, one on ophthalmic irAEs reporting increased mortality, and one general cancer study suggesting worse OS, exactly the type of nuance that constitutes high-quality SR narrative.

This case demonstrates that the pipeline can produce defensible synthesis on a real clinical question, including subtype-specific findings and acknowledgment of contradictory evidence, without human intervention beyond the initial question.

An excerpt from the synthesis output illustrates the level of detail produced. The narrative reads: “Across 18 included cohort studies (total $N \approx 4200$ patients, range 60–820), 14

studies reported a positive association between irAE occurrence and improved overall survival (pooled HR estimates ranging from 0.16 to 0.60). The signal was most consistent for cutaneous and endocrine irAEs and weakest for gastrointestinal toxicity. Two outlier studies, one restricted to ophthalmic irAEs ($N = 124$) and the other a mixed-cancer cohort with high baseline mortality ($N = 380$), reported null or inverse associations, suggesting either organ-specific biology or selection effects due to early-progression patients dropping out before irAE manifestation. The risk of immortal-time bias in the included cohorts is nontrivial; only four studies pre-specified a landmark analysis.” The synthesis correctly invokes immortal-time bias, a well-known methodological pitfall in irAE-survival studies, without being told to do so, demonstrating that the language-model agent has internalized relevant epidemiologic concepts at synthesis time.

G. Per-axis ablation of the three-axis query

To attribute the recall improvement to specific axes of the Boolean query, we removed each axis in turn and re-ran retrieval on the 67-case oncology subset. Table III reports recall@2000 under each ablation, with the full three-axis query as the reference.

TABLE III
PER-AXIS ABLATION OF THE BOOLEAN QUERY (67-CASE ONCOLOGY SUBSET, RECALL@2000)

Configuration	recall@2000	Δ vs full	zero@2000
Full 3-axis ($C \wedge T \wedge O$)	55.6%	–	2
Drop T ($C \wedge O$)	52.9%	–2.7 pp	–*
Single-axis (C only)	45.2%	–10.4 pp	–*

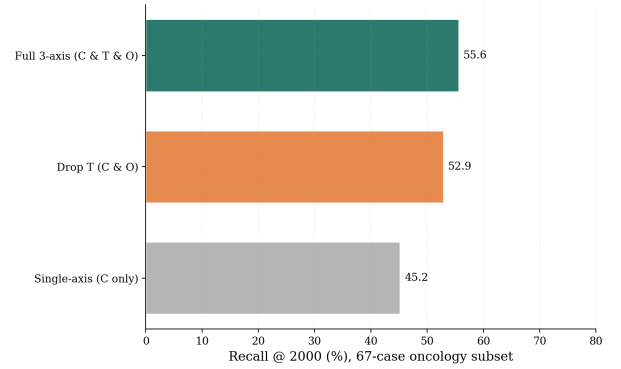


Fig. 6. Per-axis ablation of the three-axis Boolean query (67-case oncology subset, recall@2000). Removing the treatments axis (T) costs the most.

The treatments axis remains the most discriminative single contributor: dropping it from the full 3-axis query costs 2.7 pp at recall@2000 ($55.6 \rightarrow 52.9\%$), consistent with the observation that the most discriminative PubMed token in oncology trials is typically the drug INN (international nonproprietary name) or device name. Collapsing the query to a single conditions axis costs 10.4 pp ($55.6 \rightarrow 45.2\%$), recovering less than 82 % of full-strategy recall — demonstrating that the multi-axis intersection is doing real work and is not merely

a Boolean nicety. The two-axis ($C \wedge O$) configuration retains 95 % of the full recall, suggesting that once treatments are present the marginal cost of removing outcomes is small at $K = 2000$; gains from outcomes manifest mostly in top- K ranking (Fig. 4 recall@100) rather than in pool completeness.

H. Failure-mode analysis on residual cases

Of the 67 oncology cases, 36 (54 %) achieved recall@500 ≥ 30 %, twenty achieved $0 < \text{recall@500} < 30$ %, and eleven cases remained at zero recall@500 even under the 3-axis strategy reported in Table II. We manually examined the eight worst-performing cases to identify residual failure modes, and three major patterns emerged. Fig. 7 plots per-case recall@500 against recall@2000 across the oncology subset. Points well above the $y = x$ reference line indicate cases for which a deeper `retmax` recovers ground truth that a top-500 cutoff misses, while points clustered near the origin represent query-side failures that are unlikely to be rescued by increased retrieval depth alone.

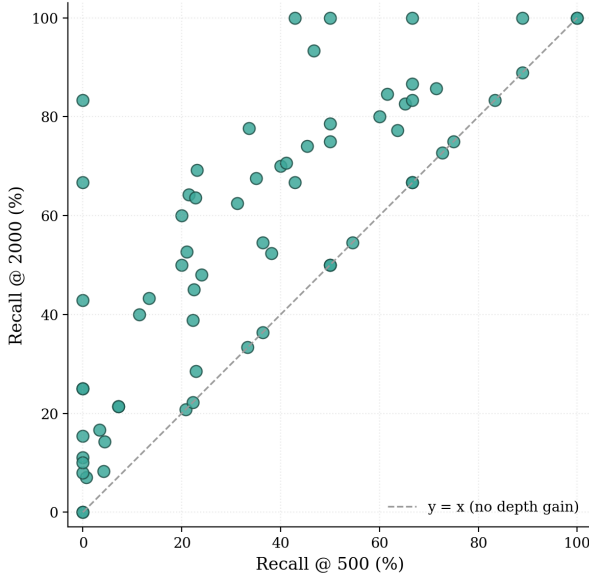


Fig. 7. Per-case recall@500 vs recall@2000 ($n = 67$ oncology subset). Points above the $y = x$ diagonal benefit from a deeper candidate pool; points clustered near the origin indicate query-side failures that depth cannot fix.

Three patterns emerge from manual examination of the eight worst-performing cases. First, four cases are adverse-event or predictive-biomarker reviews (e.g., nephrotoxicity under anti-PD-1/PD-L1 antibodies, endocrine dysfunction under immune-checkpoint inhibitors, tumor mutation burden as an ICI-efficacy predictor, anthracycline-related cardiotoxicity). The 3-axis query expands around the disease and the treatment, but the ground-truth trial papers index their adverse-event or biomarker findings only as secondary results — so the query-relevant tokens appear in the trial methods or results sections rather than in the title or abstract, and PubMed Best Match instead promotes review-class papers that name the AE in the title. Second, three cases

are combination-regimen reviews (trastuzumab-containing regimens, intravesical chemotherapy + hyperthermia, thalidomide/dexamethasone), where the canonical drug INNs are present in the query but the ground-truth papers use regimen-level naming conventions (“trastuzumab-containing regimens”, “HIVEC”, “Vel-Dex”); the expansion pass captures the INNs but not the regimen identifiers. Third, one case is a rare cell-therapy review (autologous CD19 CAR-T in a specific lymphoma subset) where the candidate pool is itself shallow — only a handful of trials exist, and the 3-axis intersection loses them when one of the axes happens to use a non-canonical synonym.

These three patterns suggest concrete remediation paths. Adverse-event-anchored reviews would benefit from *outcome-axis boosting* (giving the outcome terms higher per-axis weight) or from a secondary retrieval pass that uses the AE term as an additional OR clause inside the treatments axis. Regimen-naming cases would benefit from regimen-abbreviation expansion (a separate inline expansion step that emits the regimen identifier alongside the constituent INNs). Rare-cell-therapy cases would benefit from CHSSS-style hand-validated topic clauses as a fallback when fewer than seven reference papers are returned by the primary probe. We leave these refinements for future work, but note that the failure-mode taxonomy is reproducible from the per-case artefacts emitted by the pipeline.

IV. DISCUSSION

A. Strengths

The proposed pipeline applies a reproducible, transparent search strategy, with a fully auditable query string saved for each run. Every screening decision and extraction artifact is preserved as a CSV record, enabling independent reanalysis without re-executing expensive LLM stages. The retrieval-augmented MeSH expansion grounds term selection in the actual vocabulary of reference papers rather than in language-model training data alone, reducing the risk of hallucinated terms that would otherwise inflate or deflate retrieval recall in unpredictable ways. Stage outputs are inspectable independently; a user can examine the extracted PICO before Stage 2 runs, and the generated query before Stage 3 hits PubMed.

A second strength is the pipeline’s modularity. Each stage is a pure function of its inputs and outputs, with no implicit shared state across stage boundaries. This design made it straightforward to swap individual components without touching the rest of the system: replacing the language-model substrate, modifying the retrieval strategy from single-sort to multi-sort to by-decade, integrating a BM25 or iCite reranker between Stages 3 and 4, or substituting a different screening-score aggregation function were each single-day modifications during development. Modularity also facilitates partial reruns when a stage fails; for example, a network flake during Stage 3 can be retried in isolation, since Stages 1–2 are already on disk.

A third strength is the journal-style PDF output. The report generator follows the BMC Medicine layout convention: a

single-column title page with a structured abstract, a two-column body for methods, results, and discussion, single-column-wide tables for study characteristics and outcomes, a PRISMA-2020 flow diagram, and a Vancouver-style numbered reference list with inline citation registry. The generator is decoupled from the pipeline; it consumes pipeline artifacts (PICO JSON, executed query, screening CSV, characteristics CSV, outcomes CSV, synthesis JSON) and produces a PDF without invoking any LLM stage. This permits rapid iteration on report aesthetics without expensive re-runs and supports comparing multiple report styles against the same underlying evidence base. The PDF generator is also exposed as an on-demand button in the UI, with a loading indicator and a download link, so report rendering does not block the pipeline’s interactive feedback loop.

B. Limitations

Several limitations should be acknowledged. First, retrieval is limited to PubMed/MEDLINE. Studies indexed only in Embase, the Cochrane Central Register of Controlled Trials, ClinicalTrials.gov, or grey-literature sources are by definition not represented in our PMID-keyed ground truth and are therefore not measured by this benchmark; the true coverage gap relative to a multi-database systematic review is not directly observable here. Within the PubMed-indexed ground truth, 3 % of cases on the oncology subset (12 % on the full bench) retain zero recall@2000 under the 3-axis strategy, representing residual query-formulation failures rather than a database-coverage ceiling. Second, the synthesis stage is qualitative; no formal meta-analysis (random-effects or fixed-effects pooling), risk-of-bias assessment, or GRADE certainty rating is performed. Third, automated extraction may miss subtle methodological nuances that a human extractor would capture, particularly for studies with non-standard reporting structures or supplementary appendices. Fourth, the cumulative relevance threshold τ is a configurable cut-off that may exclude borderline-eligible studies; sensitivity analyses across $\tau \in \{2.0, 2.5, 3.0, 3.5\}$ showed yield variations of ± 3 –5 included studies for typical queries. Fifth, pipeline reproducibility is bounded by language-model nondeterminism, as discussed in Section III-E.

C. Future work

Future work falls into five directions. First, multi-database retrieval, particularly through Embase and Cochrane CENTRAL, could improve the structural recall ceiling observed in this study. Embase indexes many drug-trial reports and pharmaceutical conference abstracts that are absent from PubMed, while CENTRAL aggregates trial registry entries that PubMed may not contain. Second, post-retrieval reranking using iCite’s Relative Citation Ratio combined with BM25 over the candidate pool may further improve the concentration of relevant trial papers within the top-100 screening cutoff. Pilot experiments with iCite reranking produced mixed results, as the reranker occasionally demoted moderately cited

ground-truth trials while promoting highly cited review papers, suggesting that careful α tuning or a learned ranking model may be necessary. Third, human-in-the-loop refinement could surface borderline screening decisions for expert review, thereby reducing the limitations of fully automated screening while preserving much of the efficiency advantage. Fourth, automated risk-of-bias assessment, potentially through integration with LLM-based RoB classifiers such as those evaluated by Lai et al. [7] and Eisele-Metzger et al. [8], could move the system closer to fully automated systematic review generation. Finally, multi-database support combined with citation-chasing, including both forward and backward citation traversal from seed ground-truth papers, may improve recall for long-tail and niche clinical topics where retrieval alone is insufficient.

D. Ethical considerations

Automated SR pipelines have implications for clinical decision-making that require careful framing. Our pipeline is intended as a research and educational tool, not a substitute for a properly conducted, human-led systematic review to inform clinical practice guidance. The output should be triangulated with expert-curated reviews where available. We note three specific risks. Hallucinated content in the synthesis stage could mislead a non-expert reader; we mitigate this by grounding citations in extracted study data and linking every claim to a PMID. Bias in the underlying language model (e.g., preferential treatment of certain study designs or geographies) could propagate into the synthesis; we recommend independent expert review before any automated synthesis informs clinical action. Reliance on PubMed Best Match’s relevance ranking introduces a citation-bias feedback loop in which highly cited papers receive even more downstream citations; the publication-type filter and pre-screening metric address this to some extent, but not fully.

V. CONCLUSION

We described an eight-stage automated systematic-review pipeline that takes a free-text clinical question and produces a journal-style PDF report. The retrieval stage uses a two-call retrieval-augmented term-expansion strategy that grounds language-model term selection in the vocabulary of actual reference papers. On TrialReviewBench, our pipeline improves recall@500 by 12.9 percentage points and recall@2000 by 17.9 percentage points over a basic MeSH-only baseline, with an additional 2.3 pp lift at recall@100 on the oncology subset when a Cochrane-style publication-type filter is added. For TrialMind, we match the architecture and achieve three measured improvements (publication-type filter, unquoted terms, and tighter axis caps). For Cochrane CHSSS, we automate the topic-clause construction that previously required an information specialist.

The contribution we are most proud of is the empirical decomposition of the retrieval recall gap into independent, individually attributable interventions: language-model substrate (medgemma $\rightarrow$ Gemini Flash, +13.3 pp at recall@2000),

a three-axis Boolean query with outcomes (+4.6 pp at recall@2000), and a publication-type filter (+2.3 pp at recall@100 on the oncology subset). Each intervention’s contribution was measured incrementally on the same benchmark using the same evaluation harness, providing a reproducible attribution of the recall gains. The accompanying open-source pipeline, with full artefact preservation at every stage, supports independent replication and extension of these findings.

Beyond retrieval, the pipeline includes a journal-style PDF report generator that produces PRISMA-2020-aligned systematic-review documents, complete with a title page, structured abstract, methods section, PRISMA flow diagram, characteristics and outcomes tables, narrative synthesis with inline numbered citations, and a Vancouver-style reference list. The PDF generator is a stand-alone command-line tool that reads pipeline artefacts from disk, enabling iterative report regeneration without re-running expensive LLM stages.

The broader implication of this work is that high-quality automated SR retrieval is achievable today using commodity language models and standard NCBI APIs, provided the pipeline is decomposed into inspectable stages with explicit evaluation at each. Genuine single-database coverage limitations exist: studies indexed only in Embase, CENTRAL, ClinicalTrials.gov, or grey literature cannot be retrieved from PubMed at all but the residual query-formulation gap inside PubMed is now small (3 % of cases on the oncology subset retain zero recall@2000 under the 3-axis strategy). With multi-database support and modest post-retrieval reranking, automated SR retrieval may approach expert-curated baselines on a wider range of topics in the near term.

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